VECTORS

Term	*Definition*
Norm/Length	$\lVert v\rVert = \sqrt{\sum_{i=1}^{n} v_i^2}$ $\lVert v\rVert = \sqrt{v^T v}$ $\nabla\lVert v\rVert = \frac{1}{\lVert v\rVert} v^T$
Distance	$d(u,v) = \lVert u - v\rVert$
Scalar product	$u \bullet v = u^T v = \sum_k u_k v_k$

MATRICES

Determinant:

$$\det(a) = a \qquad \det\begin{pmatrix} a & b \\ c & d\end{pmatrix} = ad - cb$$

$$\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33}\end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

Inverting:

$$A = \begin{pmatrix} a & b \\ c & d\end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad-cb}\begin{pmatrix} d & -b \\ -c & a\end{pmatrix}$$

Transposing:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1\end{pmatrix}^T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1\end{pmatrix}, \quad \begin{pmatrix} 1 \\ 4 \\ 5\end{pmatrix}^T = (1 \;\; 4 \;\; 5)$$

Matrix multiplication:

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7\end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9\end{pmatrix}$$

$$= \begin{pmatrix} (2\cdot 6 + 3\cdot 1 + 1\cdot 8) & (2\cdot 4 + 3\cdot 0 + 1\cdot 9) \\ (4\cdot 6 + -1\cdot 1 + 7\cdot 8) & (4\cdot 4 + -1\cdot 0 + 7\cdot 9)\end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79\end{pmatrix}$$

INDICES

$$A \in \mathbb{R}^{n\times m}, B \in \mathbb{R}^{m\times r}, A\cdot B \in \mathbb{R}^{n\times r}$$

$$(A\cdot B)_{ij} = \sum_{k=1}^{m} A_{ik}B_{kj} = \sum_k A_{ik}B_{kj}$$

$$((A\cdot B)^T)_{ij} = (A\cdot B)_{ji} = \sum_k A_{jk}B_{ik}$$

$$a^T \cdot B \cdot b = \sum_{ij} (a^T)_i B_{ij} b_j = \sum_{ij} a_i B_{ij} b_j$$

$$(B\cdot a)(C\cdot a) = \sum_i (B\cdot a)_i (C\cdot a)_i$$

KRONECKER DELTA

$$\delta_{ij} = (\vec{e}_i)_j = \begin{cases} 1 & , i = j \\ 0 & , \text{otherwise}\end{cases}$$

$$(E_{rst})_{ijk} = \delta_{ri}\delta_{sj}\delta_{tk}$$

BASIS VECTORS

$$\mathbb{R}^3$$

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0\end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0\end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1\end{pmatrix}$$

UNIT MATRICES

$$\mathbb{R}^2$$

$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0\end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0\end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0\end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1\end{pmatrix}$$

AFFINE TRANSFORMATIONS

$$A_{a,M}(x) = a + M\cdot x$$

$$L_A\left(\vec{0}\right) = \vec{0}$$

DERIVATIVES

1	$\to 0$	x	$\to 1$
x^2	$\to 2x$	x^n	$\to nx^{n-1}$
$\frac{1}{x}$	$\to -\frac{1}{x^2}$	$\sqrt{x}$	$\to \frac{1}{2\sqrt{x}}$
e^x	$\to e^x$	e^{-x}	$\to -e^{-x}$
$\ln(x)$	$\to \frac{1}{x}$ for $x > 0$		
a^x	$\to \ln(a)\cdot a^x$ for $a > 0, a \neq 1$	$\ln(y\cdot x)$	$\to \frac{1}{x}$ for $x > 0$
$\sin(x)$	$\to \cos(x)$	$\log_b(x)$	$\to \frac{1}{\ln(b)\cdot x}$
$\cos(x)$	$\to -\sin(x)$	$\sin(2x)$	$\to 2\cos(2x)$
$\tan(x)$	$\to 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$	$\cos(ax)$	$\to -a\sin(ax)$
$\cot(x)$	$\to -\frac{1}{\sin^2(x)}$	$\text{arccot}(x)$	$\to -\frac{1}{1+x^2}$
$\arccos(x)$	$\to -\frac{1}{\sqrt{1-x^2}}$	$\arcsin(x)$	$\to \frac{1}{\sqrt{1-x^2}}$
		$\arctan(x)$	$\to \frac{1}{1+x^2}$

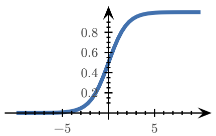
Term	*Definition*
Addition	$(f(x) + g(x))' = f'(x) + g'(x)$
Multiplication	$(\alpha\cdot f(x))' = \alpha\cdot f'(x)$
Product rule	$(f\cdot g)' = f'\cdot g + f\cdot g'$ $(f\cdot g\cdot h)' = f'\cdot g\cdot h + f\cdot g'\cdot h + f\cdot g\cdot h'$
Chain rule	$(f(g(x)))' = f'(g(x))\cdot g'(x)$
Quotient rule	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

FUNCTIONS OF SEVERAL VARIABLES

Term	*Definition*
Real valued function	$R \subset \mathbb{R}$
Vector valued function	$R \subset \mathbb{R}^m$
A function of n variables	$D \subset \mathbb{R}^n$
Softmax function	*TODO:*
RSS	$\text{RSS} = \sum_i (y_i - f(x_i))^2$
n-dimensional curve	$c : \mathbb{R} \to \mathbb{R}^n$
n-dimensional hyper-surface	$s : \mathbb{R}^n \to \mathbb{R}$
Reparametrization	$c(t) \to d(s) = c(h(s))$

Sigmoid function:

$$\sigma(x) : \mathbb{R} \to [0;1] = \frac{1}{1 + e^{-x}}$$



DERIVATIVES

Function	*Derivative*
$f : \begin{cases} \mathbb{R} \to \mathbb{R}^n \\ x \mapsto y\end{cases}$	$f' : \begin{cases} \mathbb{R} \to \mathbb{R}^n \\ x \mapsto y\end{cases} = \begin{pmatrix} \frac{d}{dx}f_1(x) \\ \frac{d}{dx}f_2(x) \\ \vdots \\ \frac{d}{dx}f_n(x)\end{pmatrix}$
$f : \begin{cases} \mathbb{R}^n \to \mathbb{R} \\ x \mapsto y\end{cases}$	$f' : \begin{cases} \mathbb{R}^n \to \mathbb{R}^{1\times n} \\ x \mapsto y\end{cases} = \nabla f$ $= \left(\frac{\partial}{\partial x_1}f(x), \frac{\partial}{\partial x_2}f(x), \dots, \frac{\partial}{\partial x_n}f(x)\right)$
$f : \begin{cases} \mathbb{R}^n \to \mathbb{R}^m \\ x \mapsto y\end{cases}$	$f' : \begin{cases} \mathbb{R}^n \to \mathbb{R}^{m\times n} \\ x \mapsto y\end{cases} = J_f = \frac{\partial f}{\partial x}\Big\vert_{x=x_0}$ $= \begin{pmatrix} \nabla f_1(x) \\ \vdots \\ \nabla f_m(x)\end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1}f_1(x) & \dots & \frac{\partial}{\partial x_n}f_1(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1}f_m(x) & \dots & \frac{\partial}{\partial x_n}f_m(x)\end{pmatrix}$

PARTIAL VS TOTAL DERIVATIVE

TODO:
example FS2024

$$f(x,y) = x^2 + 2y$$

$$\frac{\partial f}{\partial x} = 2x$$

$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\cdot\frac{dy}{dx} = 2x + 2\cdot\frac{dy}{dx} \mid (\text{y dependent on x})$$

GRADIENTS

Common gradients:

$$f(x) = \lVert x\rVert^2 \qquad \Rightarrow \nabla f = 2x$$

$$f(x) = a^T x + c \qquad \Rightarrow \nabla f = a$$

$$f(x) = x^T A x \qquad \Rightarrow \nabla f = (A + A^T)x$$

$$f(x) = \frac{1}{2}x^T A x + b^T x + c \Rightarrow \nabla f = \frac{A + A^T}{2}x + b$$

$$= Ax + b \text{ if } A^T = A$$

$$f(x) = g(Ax + b) \qquad \Rightarrow \nabla f = A^T\nabla g(Ax + b)$$

$$g : \mathbb{R}^m \to \mathbb{R}$$

JACOBIAN MATRIX

$$f : \mathbb{R}^2 \to \mathbb{R}^2, \; J_f(x,y) = \begin{pmatrix} \partial x_1 & \partial y_1 \\ \partial x_2 & \partial y_2\end{pmatrix}$$

Linearisation:

$$f : \mathbb{R}^n \to \mathbb{R}^m, \; x_0 \in \mathbb{R}^n$$

$$L_f(x) = f(x_0) + J_f(x_0)\cdot(x - x_0)$$

Chain rule:

$$f : \mathbb{R}^n \to \mathbb{R}^m, \; g : \mathbb{R}^m \to \mathbb{R}^k, \; h : \mathbb{R}^n \to \mathbb{R}^k = g\circ f$$

$$J_h = J_g(f(x))\cdot J_f(x)$$

Common jacobians:

$$f(x) = a + M\cdot x \Rightarrow J_f = M$$

$$f(x) = x \qquad \Rightarrow J_f = \mathbb{1}$$

$$f(x) = \lVert x\rVert^2 \qquad \Rightarrow J_f = (2x)^T$$

TODO:
J_x for softmax $s_i = \frac{e^{x_i}}{\Sigma} e^{x_k} : J_{ij} = s_{i(\delta_{ij} - s_j)}$

HYPER-SURFACES

Chain rule:

$$f : \mathbb{R}^n \to \mathbb{R}^m, \; g : \mathbb{R}^m \to \mathbb{R}, \; h : \mathbb{R}^n \to \mathbb{R} = g\circ f$$

$$\nabla h(x) = \nabla g(f(x)) = \nabla g(f)\vert_{f=f(x)}\cdot\frac{\partial}{\partial x}f(x)$$

Differentiation properties:

$$f : \mathbb{R}^n \to \mathbb{R}, \; g : \mathbb{R}^n \to \mathbb{R}$$

$$\nabla(f + g) = \nabla f + \nabla g$$

$$\nabla(f - g) = \nabla f - \nabla g$$

$$\nabla(f\cdot g) = g\nabla f + f\nabla g$$

$$\nabla\frac{f}{g} = \frac{g\nabla f - f\nabla g}{g^2}$$

COMMUTATIVE DIAGRAMS

$f : D \to M$
$g : N \to R$
$h : D \to R$
$= g\circ f$

$$\begin{tikzcd} D \arrow[r, "f"] \arrow[rr, bend left, "g\circ f"] & M \arrow[r, "g"] & R \end{tikzcd}$$

GEOMETRY OF HYPER-SURFACES

$$f : \mathbb{R}^2 \to \mathbb{R}$$

Tangent space: $\alpha, \beta \in \mathbb{R}, T =$

$$\left\{\begin{pmatrix} x_0 \\ y_0 \\ f(x_0,y_0)\end{pmatrix} + \alpha\begin{pmatrix} 1 \\ 0 \\ \partial_x f(x_0,y_0)\end{pmatrix} + \beta\begin{pmatrix} 0 \\ 1 \\ \partial_y f(x_0,y_0)\end{pmatrix}\right\}$$

Graph of a linearisation:

$$l(x) = f(x_0) + \nabla f(x_0)\cdot(x - x_0) = T_{p_0}$$

SECOND DERIVATIVE

$$\frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}(f(x,y))\right) = \frac{\partial^2 f(x,y)}{\partial x^2} = \partial_{xx}f(x,y)$$

HESSIAN MATRIX

$$f : \mathbb{R}^n \to \mathbb{R}$$

$$H(x,y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy}\end{pmatrix} = J(\nabla f)$$

$$(H(x))_{ij} = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j}f(x)$$

$$\partial_i\partial_j f = \partial_j\partial_i f$$

Common hessians:

$$f(x) = \frac{1}{2}x^T A x + b^T x + c \Rightarrow H_f = \frac{A + A^T}{2}$$

$$f(x) = g(Ax + b) \qquad \Rightarrow H_f = A^T H_g(Ax + b)A$$

$$g : \mathbb{R}^m \to \mathbb{R}$$

EXTREMAL POINTS

Let $z = f(c(t))$, c a curve passing the stationary point $c_0 = (x_0, y_0)$ at $t = t_0$ and

$$\frac{d^2 z}{dt^2} = \underbrace{(\dot{c}_1(t_0), \dot{c}_2(t_0))}_{v^T}\cdot\underbrace{\begin{pmatrix} f_{xx}(c_0) & f_{xy}(c_0) \\ f_{yx}(c_0) & f_{yy}(c_0)\end{pmatrix}}_{H}\cdot\underbrace{\begin{pmatrix} \dot{c}_1(t_0) \\ \dot{c}_2(t_0)\end{pmatrix}}_{v}$$

1) If $\forall v \in \mathbb{R}^2 \setminus \left\{\vec{0}\right\}(v^T H v > 0)$, then $\frac{d^2 z}{dt^2} > 0$ defines a ***local minimum***
2) If $\forall v \in \mathbb{R}^2 \setminus \left\{\vec{0}\right\}(v^T H v < 0)$, then $\frac{d^2 z}{dt^2} < 0$ defines a ***local maximum***
3) If $\forall v \in \mathbb{R}^2 \setminus \left\{\vec{0}\right\}(v^T H v < 0 \vee v^T H v > 0)$, the point $c_0 = (x_0, y_0)$ defines a ***saddle point***

TODO:

$$(J_u)_{ij} = \frac{\partial u_i}{\partial x_j}, (H_g)_{ij} = \frac{\partial^2 g}{\partial x_i \partial x_j}$$

$$\partial_j(u_i v_i) = (\partial_j u_i)v_i + u_i(\partial_j v_i)$$

$$(v_1, v_2, v_3)\cdot\begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33}\end{pmatrix}\cdot\begin{pmatrix} v_1 \\ v_2 \\ v_3\end{pmatrix}$$

$$= (v_1, v_2, v_3)\begin{pmatrix} v_1 h_{11} + v_2 h_{12} + v_3 h_{13} \\ v_1 h_{21} + v_2 h_{22} + v_3 h_{23} \\ v_1 h_{31} + v_2 h_{32} + v_3 h_{33}\end{pmatrix}$$

$$= v_1^2 h_{11} + v_1 v_2 h_{12} + v_1 v_3 h_{13}$$
$$+ v_1 v_2 h_{21} + v_2^2 h_{22} + v_3 v_2 h_{23}$$
$$+ v_1 v_3 h_{31} + v_2 v_3 h_{32} + v_3^2 h_{33}$$
$$= v_1^2 h_{11} + v_2^2 h_{22} + v_3^2 h_{33} + 2v_1 v_2 h_{12} + 2v_2 v_3 h_{23} + 2v_1 v_3 h_{13}$$

Technique of completing squares

$$\underbrace{x^2}_{\sqrt{\cdot}} - \underbrace{4xy}_{\div 2x} + 1$$

$$= (x - 2y)^2 - (-2y)^2 + 1$$

$$= (x - 2y)^2 - 4y^2 + 1$$

MISC

$$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

TRIGONOMETRY

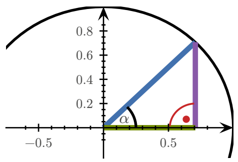
x	0	$30° = \frac{\pi}{6}$	$45° = \frac{\pi}{4}$	$60° = \frac{\pi}{3}$	$90° = \frac{\pi}{2}$
$\sin(x)$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos(x)$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan(x)$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

$$\sin(2x) = 2\sin(x)\cos(x)$$

$$\sin^2(x) + \cos^2(x) = 1$$

$1, \sin(\alpha), \cos(\alpha)$



GRADIENT DESCENT / NEWTON'S METHOD

$$GD : x_{i+1} = x_i - \gamma\cdot\nabla f(x_i)$$

$$N : x_{i+1} = x_i - \gamma\cdot\left(H_{f(x_i)}\right)^{-1}\cdot\nabla f(x_i)$$

$$\gamma = \text{step size}$$

$$\text{termination criterion } \varepsilon > \lvert x_{i+1} - x_i\rvert$$

EXAMPLES

TODO:

WORKING WITH INDICES

$x, y \in \mathbb{R}^n,\ f : \mathbb{R}^n \to \mathbb{R}$

$$f(x) = (x-y)^T(3x-y) = \sum_{i=1}^{n}(x_i - y_i)(3x_i - y_i)$$

$\partial x_i f(x) = (3x_i - y_i) + 3(x_i - y_i)$

$$f(x) = \ln(x_1 \cdot x_2 \cdot ... \cdot x_n) - |x|^2 = \sum_{i=1}^{n}(\ln(x_i) - x_i^2)$$

$\partial x_i f(x) = \frac{1}{x_i} - 2x_i$